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Seasonal and spatial variations of phytoplankton community in Tilaiya Dam, Jharkhand in relation to anthropogenic influences

Soni Kumari[a]*, Ajay Kumar Choudhary[b] & Prasanjit Mukherjee[c]
[a]University Department of Environmental Studies, DSPM University, Ranchi, Jharkhand, India
[b]University Department of Zoology, DSPM University, Ranchi, Jharkhand, India.
[c]Department of Botany, K.O.College, Gumla, Ranchi University, Ranchi, Jharkhand, India

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Abstract- Phytoplankton forms the base of aquatic food webs and serves as sensitive indicators of water quality. The present study investigates the seasonal and spatial variation of phytoplankton communities in Tilaiya Dam, Jharkhand, over two consecutive years (2021-2022 and 2022-2023). Four sampling sites were selected based on varying anthropogenic influences, including agricultural runoff, cage culture activity, human encroachment, and vehicular pollution. A total of four major phytoplankton groups were identified: Cyanophyceae, Chlorophyceae, Bacillariophyceae, and Euglenophyceae. Among these, Cyanophyceae dominated throughout the study period, with *Microcystis* sp. showing the highest abundance, particularly during winter months. Chlorophyceae exhibited moderate diversity, while diatoms showed seasonal peaks during cooler periods. Euglenophyceae were recorded in lower numbers but indicated organic enrichment in certain areas. The results reveal a gradual increase in phytoplankton density in the second year, suggesting enhanced nutrient loading. Seasonal trends indicated maximum abundance during winter and minimum during monsoon. The study highlights the impact of anthropogenic activities on phytoplankton structure and emphasizes the need for regular monitoring and sustainable management of freshwater ecosystems.

Keywords: Phytoplankton, Tilaiya Dam, Seasonal variation, Cyanophyceae, Eutrophication, Anthropogenic impact

INTRODUCTION

Phytoplanktons are microscopic, photosynthetic organisms that remain suspended in aquatic systems and form the base of the food web. Through photosynthesis, they convert inorganic substances into organic matter, thereby supporting higher trophic levels such as zooplankton and fish. They also contribute significantly to global oxygen production and play an important role in regulating nutrient cycles in both freshwater and marine environments.[1,2]

*Corresponding author :
Phone : 7488119520
E-mail : sonikumarijan8@gmail.com

The structure and distribution of phytoplankton communities are highly responsive to environmental conditions. Factors such as nutrient concentration, temperature, light availability, and hydrological changes strongly influence their growth and composition. Anthropogenic activities including agricultural runoff, aquaculture practices, urban discharge, and surface runoff can significantly alter these parameters, often leading to shifts in community structure.[3,4] Due to their rapid response to such changes, phytoplankton are widely used as reliable indicators of water quality and trophic status in limnological studies.[5,6]

In recent years, increasing human pressure on freshwater ecosystems has resulted in nutrient enrichment and ecological imbalance, particularly in reservoir systems. Excessive input of nitrogen and phosphorus often promotes the proliferation of cyanobacteria, leading to eutrophication and deterioration of water quality. Understanding the dynamics of phytoplankton communities under such influences is therefore essential for assessing ecosystem health and sustainability.

The present study focuses on evaluating the diversity, seasonal variation, and spatial distribution of phytoplankton in relation to anthropogenic disturbances. Special emphasis is placed on understanding how different environmental pressures influence phytoplankton composition and abundance over time.

MATERIALS & METHODS

Study Area and Sampling Sites

The present investigation was carried out in Tilaiya Dam, located in the Koderma district of Jharkhand, India. Geographically, the reservoir lies at approximately 24°20' 38.8" N latitude and 85°25'56.201" E longitude. Constructed across the Barakar River under the Damodar Valley Corporation (DVC), the dam forms part of an important multipurpose river valley system. The reservoir extends over an area of nearly 60 km² and receives an average annual rainfall of about 127 cm. The impounded region has developed into a productive wetland ecosystem supporting diverse biological communities and providing essential resources for local populations.

To estimate the influence of varying environmental conditions and anthropogenic activities on phytoplankton distribution, four distinct sampling sites were selected within the reservoir. Each site represents a specific type of land-use pattern and degree of human interference.

- **Site I (Agricultural Area):** This zone is primarily influenced by agricultural runoff, receiving nutrient inputs such as nitrogen and phosphorus from fertilizers.
- **Site II (Cage Culture Area):** Characterized by aquaculture activities, this site is enriched with nutrients derived from fish feed and metabolic wastes.
- **Site III (Encroachment Zone):** This area is subjected to intense human interference, including domestic activities and waste discharge, leading to organic enrichment.
- **Site IV (Bridge Area):** Located near a roadway, this site is affected by vehicular runoff and increased suspended solids.

The selection of these sites enabled a comparative assessment of phytoplankton composition in relation to different anthropogenic pressures and environmental gradients within the reservoir.[7]

Sampling and Analysis

Phytoplankton sampling was conducted at monthly intervals over a two-year period from November 2021 to October 2023 to capture both seasonal and annual variations in community structure. Samples were collected from all selected sites using standard plankton collection techniques. Surface water was filtered through a plankton net (mesh size ~20–25 µm) to concentrate phytoplankton, and the collected samples were transferred into clean, labeled containers.

After collection, the samples were preserved using Lugol's iodine solution to prevent decomposition and to maintain cellular integrity for further analysis. In the laboratory, the preserved samples were allowed to settle, and concentrated aliquots were used for microscopic examination.

Phytoplankton identification was carried out up to the genus level using standard taxonomic keys and monographs, including those of Prescott (1982)[8] and Bellinger and Sigee (2010)[2]. Morphological characteristics such as cell shape, size, colony formation, and pigmentation were used as primary criteria for identification.

Quantitative estimation of phytoplankton was performed using a Sedgwick-Rafter counting chamber under a compound microscope. The number of individuals was counted systematically across multiple fields to ensure accuracy, and the results were expressed in terms of units per liter. Care was taken to minimize counting errors by conducting repeated observations and maintaining uniform sampling procedures.

This approach ensured a systematic and dependable evaluation of phytoplankton diversity, abundance, and seasonal variation under differing environmental conditions within the reservoir.

Composition and Abundance of Phytoplankton

The phytoplankton community of Tilaiya Dam during the study period was represented by four major groups:

Cyanophyceae, Chlorophyceae, Bacillariophyceae, and Euglenophyceae. Among these, Cyanophyceae emerged as the most dominant group across both years, followed by Chlorophyceae, Euglenophyceae and Bacillariophyceae.

Table 1. Group-wise mean abundance (units) of phytoplankton (2021–2023)

Group	2021–2022 (Mean)	2022–2023 (Mean)	Trend
Cyanophyceae	~150–200	~170–220	Increasing
Chlorophyceae	~60–80	~70–90	Slight rise
Bacillariophyceae	~40–60	~50–70	Moderate rise
Euglenophyceae	~15–30	~20–35	Slight rise

Cyanophyceae contributed nearly half of the total phytoplankton population, clearly indicating their ecological dominance (table 1). The consistent increase in their abundance during 2022–2023 suggests enhanced nutrient availability, particularly nitrogen and phosphorus. Chlorophyceae maintained a moderate but stable presence, reflecting relatively balanced environmental conditions. Bacillariophyceae showed moderate abundance with seasonal sensitivity, while Euglenophyceae remained the least abundant but showed a noticeable increase in the second year, possibly indicating localized organic enrichment.

Seasonal Variation

Phytoplankton abundance exhibited marked seasonal fluctuations, reflecting the influence of climatic and hydrological factors.

Table 2. Seasonal variation in total phytoplankton abundance (mean units)

Season	2021–2022	2022–2023	Observation
Winter	High	Higher	Peak productivity
Summer	Moderate	Moderate	Stable growth
Monsoon	Low	Low	Decline due to dilution
Post-monsoon	Moderate	Moderate	Recovery phase

It clearly indicates that winter season supported maximum phytoplankton growth in both years (table 2). This may be attributed to stable water conditions, increased transparency, and optimal nutrient availability, which together favor photosynthetic activity. Monsoon months showed a sharp decline in phytoplankton density due to increased turbidity, reduced light penetration, and flushing effects caused by heavy rainfall. Summer and post-monsoon seasons represented transitional phases, where moderate abundance reflects gradual ecological adjustment.

Dominant Species Contribution

A few dominant taxa played a significant role in shaping the overall phytoplankton community structure.

Table 3. Dominant phytoplankton taxa and peak abundance (units)

Taxa	2021–2022 Peak	2022–2023 Peak	Month of Peak
Microcystis sp.	500	520	February
Anabaena sp.	150	155	June
Oscillatoria sp.	95	100	April
Pediastrum sp.	85	88	June
Navicula sp.	95	100	February
Euglena sp.	40	45	February

Microcystis sp. was the most dominant species, reaching peak abundance during February in both years (table 3). This indicates its strong adaptability to favorable winter conditions and nutrient-rich environments. Other cyanobacterial taxa such as *Anabaena* and *Oscillatoria* also showed significant presence, reinforcing the dominance of Cyanophyceae. Diatoms like *Navicula* exhibited winter peaks, suggesting their preference for cooler and well-oxygenated conditions. The presence of *Euglena* in measurable quantities further reflects organic enrichment in certain zones.

Group-wise Seasonal Trends

The seasonal behavior of different phytoplankton groups provides insight into their ecological preferences.

Table 4. Seasonal dominance of phytoplankton groups

Group	Winter	Summer	Monsoon	Post-monsoon
Cyanophyceae	Very high	High	Moderate	High
Chlorophyceae	Moderate	Moderate	Low	Moderate
Bacillariophyceae	High	Moderate	Low	Moderate
Euglenophyceae	Low	Low	Moderate	Low

Cyanophyceae maintained dominance throughout the year, with particularly high abundance during winter and summer (table 4). This persistent dominance indicates eutrophic conditions in the reservoir. Bacillariophyceae showed a clear preference for winter, likely due to favorable temperature and light conditions, while their decline during monsoon is linked to increased turbidity. Chlorophyceae exhibited relatively stable distribution, whereas Euglenophyceae showed slight increases during monsoon, possibly due to organic matter accumulation.

Spatial Variation

The distribution of phytoplankton varied significantly across different sampling sites, reflecting the impact of local environmental conditions and anthropogenic activities.

The agricultural site exhibited strong cyanobacterial dominance due to nutrient enrichment from fertilizer runoff (table 5). The cage culture site supported relatively higher diversity, likely due to continuous nutrient input from aquaculture activities. The encroachment site recorded the highest phytoplankton density but reduced diversity, indicating ecological stress and organic pollution. The bridge site was characterized by the dominance of pollution-tolerant species, reflecting the impact of vehicular emissions and suspended solids.

Table 5. Site-wise characteristics and phytoplankton response

Site	Environmental Condition	Phytoplankton Response
Agricultural Site	Fertilizer runoff	High cyanobacterial dominance
Cage Culture Site	Nutrient input (fish culture)	High diversity
Encroachment Site	Organic pollution	High density, low diversity
Bridge Site	Vehicular runoff	Pollution-tolerant species dominant

Inter-annual Comparison

A comparison between the two study years revealed a gradual shift in phytoplankton dynamics.

Table 6. Inter-annual comparison of key indicators

Parameter	2021–2022	2022–2023	Trend
Total phytoplankton density	Moderate	Higher	Increasing
Cyanophyceae dominance	High	Higher	Increasing
Species diversity	Moderate	Slightly reduced	Declining
Pollution indicators	Moderate	Higher	Increasing

A clear increase in total phytoplankton density during 2022–2023 (table 6). This rise was primarily driven by Cyanophyceae, suggesting enhanced nutrient loading and progressive eutrophication. At the same time, a slight decline in species diversity was observed, which may be due to competitive exclusion by dominant taxa such as *Microcystis*. The increase in pollution-indicator species further supports the inference of deteriorating water quality over time.

DISCUSSION

The present study demonstrates that the phytoplankton community of Tilaiya Dam is predominantly composed of Cyanophyceae, with clear seasonal and spatial variations influenced by environmental conditions and anthropogenic pressures. The dominance of cyanobacteria, particularly *Microcystis*, *Anabaena*, and *Oscillatoria*, is a well-established feature of nutrient-enriched freshwater systems and is widely associated with eutrophic conditions driven by elevated nitrogen and phosphorus inputs.[9-11] The increasing trend of Cyanophyceae observed during the second year of the study suggests progressive nutrient loading, likely resulting from agricultural runoff, aquaculture activities, and domestic waste inputs within the catchment area.

Seasonal variation in phytoplankton abundance followed a distinct pattern, with maximum density recorded during winter and minimum during the monsoon period. Such seasonal dynamics are consistent with classical limnological observations, where stable thermal stratification, improved water transparency, and optimal nutrient availability during winter favor phytoplankton growth.[12] Similar seasonal peaks have been reported in freshwater ecosystems under comparable climatic conditions.[13] The decline during monsoon can be attributed to increased turbidity, reduced light penetration, and flushing effects, which collectively limit photosynthetic activity and reduce phytoplankton standing crop.[14]

The pronounced dominance of *Microcystis* during winter months is particularly significant, as this genus is globally recognized for its role in harmful algal blooms and its strong association with eutrophic waters.[15-17] The observed increase in its peak abundance over the study period indicates a shift toward more nutrient-rich conditions and raises concerns regarding potential bloom formation and associated ecological risks. Such cyanobacterial proliferation is often linked to anthropogenic nutrient enrichment and has been widely documented in freshwater reservoirs.[18]

Chlorophyceae exhibited moderate and relatively stable distribution throughout the study period, suggesting their adaptability to a range of environmental conditions. This stability reflects their ecological flexibility and ability to persist under both mesotrophic and moderately eutrophic conditions, as also reported in earlier studies on phytoplankton community structure.[16] Bacillariophyceae (diatoms) showed clear seasonal peaks during winter, which can be attributed to their preference for cooler temperatures, higher dissolved oxygen levels, and lower turbidity.[19] Their decline during monsoon further supports

the role of suspended solids and reduced light availability in limiting diatom growth.

Euglenophyceae, although least abundant, were consistently present, particularly in areas subjected to organic pollution. The occurrence of genera such as *Euglena* and *Phacus* is commonly associated with organically enriched environments and serves as an indicator of deteriorating water quality.[5] Their localized increase in the encroachment zone of the reservoir highlights the impact of domestic waste discharge and organic loading on phytoplankton composition.

Spatial variation in phytoplankton distribution further emphasizes the influence of anthropogenic activities on community structure. The agricultural site showed strong cyanobacterial dominance due to nutrient inflow from fertilizers, a phenomenon widely recognized as a key driver of algal blooms in freshwater systems.[10] The cage culture site supported relatively higher diversity, likely due to continuous nutrient input from fish feed and excreta, creating a nutrient-rich yet dynamic environment. In contrast, the encroachment site exhibited high phytoplankton density but reduced diversity, indicating ecological stress and competitive dominance under polluted conditions. Such patterns are characteristic of disturbed ecosystems where a few tolerant species outcompete others.[20]

The inter-annual comparison revealed a clear increase in phytoplankton density accompanied by a slight decline in species diversity, indicating a shift toward eutrophic conditions. This inverse relationship between biomass and diversity is a well-documented ecological response to nutrient enrichment, where increased nutrient availability favors the proliferation of a few dominant taxa at the expense of overall community diversity.[21] The rising dominance of Cyanophyceae and pollution-indicator species in the present study strongly suggests gradual ecological degradation of the reservoir.

The phytoplankton dynamics observed in Tilaiya Dam reflect a complex interaction between natural seasonal processes and anthropogenic influences. The increasing trend of cyanobacterial dominance and nutrient enrichment highlights the need for continuous monitoring and effective management strategies to control nutrient inputs and prevent further deterioration of water quality.

CONCLUSION

The study demonstrates that phytoplankton dynamics in Tilaiya Dam are strongly influenced by seasonal changes and anthropogenic activities. The consistent dominance of Cyanophyceae reflects eutrophic conditions and highlights the impact of nutrient enrichment.

There is a noticeable increase in phytoplankton density in the second year, suggesting gradual deterioration of water quality. Effective management strategies, including control of nutrient input and monitoring of anthropogenic activities, are essential to maintain ecological balance in the reservoir.

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